How Fission Gas Pressurised Pores Drive High Burnup Fuel Fragmentation: A Micromechanics-Informed Pore Population Model in SCIANTIX

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Abstract

Fine fragmentation of high burnup nuclear fuel is a licensing issue in the United States, where the economics of 24-month cycles would push peak rod burnups towards 75 GWd/tU, past the fragmentation threshold and beyond the 62 GWd/tU to which pressurised water reactors are currently licensed. European regulators judge that such fragmentation does not arise at present burnups and core loadings, yet the phenomenon itself was discovered and characterised in European facilities, such as Halden and Studsvik, whose programmes are the experimental basis for any future burnup extension.

Fine fragmentation is attributed to the fission gas stored in overpressurised micrometric pores of the high burnup structure at the pellet rim. During a loss-of-coolant accident, as the fuel heats up and the external mechanical restraint on the pellet vanishes, the pore pressure fractures the surrounding matrix into fragments of tens of micrometres and the gas is suddenly released.

Safety analyses still predict the onset with empirical burnup-temperature thresholds, which resolve neither the fission-gas-driven mechanism nor the microstructural evolution before the transient. Mechanistic rupture criteria based on pore overpressure are available in fuel performance codes, but simplify the pore-pressure state. The latter criteria either prescribe the pore population, assuming that every pore reached the dislocation-punching pressure during base irradiation, or consider a single mean pore, and they neglect the interaction between pores. The distribution of pore size, gas content and local stress, which determines where and when rupture starts, is therefore not calculated and not informed by the irradiation conditions.

Here we show that mesoscale micromechanics calculations coupled with fission gas behaviour models supply this description within engineering fuel performance codes.

The model is implemented in SCIANTIX, a open-source code developed at Politecnico di Milano dedicated to fission gas behaviour. SCIANTIX describes the formation of the high burnup structure together with the mean volume of its pores, their mean gas content, and the variance of that content. From these quantities the model builds a population of pore size classes, each with its own pressure. The local stress varies from pore to pore with the size and pressure distributions and with the random arrangement of the neighbours. This spread is computed with MMM, a mesoscale finite element code that solves the elastic problem on a periodic representative volume element of randomly placed pressurised spherical pores. The resulting stress concentration factors depend on a few dimensionless numbers only, so a surrogate model fitted to the MMM simulations provides them to SCIANTIX. The model is assessed against the annealing test of a high burnup fuel disc, a single-effect test that isolates the pore-pressure mechanism. The pores break progressively as the temperature rises, the smallest classes first, and within a class the

pores with the highest local stress first. This progressive rupture is consistent with the measured release and with earlier pore-distribution models. Integral loss-of-coolant tests of the Halden IFA-650 series are also used to verify the fine fragmentation threshold, although cladding strain, ballooning, pellet-cladding contact and fuel relocation complicate their description.

Since SCIANTIX is already coupled to several fuel performance codes, this model offers the possibility to incorporate a micromechanically-informed rupture criterion directly into engineering-scale fuel safety analyses.

Keywords: fragmentation, high burnup structure, fission gas release.